Finally,

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(\cos(xy^2)2xy + 5e^{x^3+5y}\right)$$

$$= -\sin(xy^2)y^2 2xy + \cos(xy^2)2y + 5e^{x^3+5y}3x^2$$

$$= 2y\cos(xy^2) - 2xy^3\sin(xy^2) + 15x^2e^{x^3+5y}.$$

Note that we have first used sum rule, then in the next step we have used chain rule. In the third step, product rule is used. Also, we see that $f_{xy} = f_{yx}$. Is it a coincidence? or is it always true? Actually, there are functions for which $f_{xy} \neq f_{yx}$ at some points. The following theorem gives conditions under which $f_{xy} = f_{yx}$.

Theorem 8.1 (Clairaut's Theorem)

Suppose that $A = \{(x, y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2$, $F : A \to \mathbb{R}$. If f_{xy} and f_{yx} exist in A are continuous in A, then $f_{xy} = f_{yx}$ in A.

We omit the discussion on the proof at this stage.

Example 8.14

Let $w(x, y) = xy + \frac{e^y}{y^2+1}$ for all $(x, y) \in \mathbb{R}^2$. Calculate $\frac{\partial^2 w}{\partial y \partial x}$ and $\frac{\partial^2 w}{\partial x \partial y}$.

Solution

First we calculate $\frac{\partial w}{\partial x}(x, y) = \frac{\partial(xy)}{\partial x} + \frac{\partial\left(\frac{e^y}{y^2+1}\right)}{\partial x}$.

This gives $\frac{\partial w}{\partial x}(x, y) = y + 0$ and hence $\frac{\partial^2 w}{\partial y \partial x}(x, y) = 1$. On the other hand,

$$\frac{\partial w}{\partial y}(x, y) = \frac{\partial(xy)}{\partial y} + \frac{\partial\left(\frac{e^y}{y^2+1}\right)}{\partial y}.$$

$$= x + \frac{(y^2+1)e^y - e^y 2y}{(y^2+1)^2}.$$

Hence $\frac{\partial^2 w}{\partial x \partial y}(x, y) = 1$.

Definition 8.9

Let $A = \{(x, y) \mid a < x < b, c < y < d\} \subset \mathbb{R}^2$. A function $u : A \to \mathbb{R}^2$ is said to be **harmonic** in A if it satisfies $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0, \forall (x, y) \in A$. This equation is called **Laplace's** equation.

Laplace's equation occurs in the study of many natural phenomena like heat conduction, electrostatic field, fluid flows etc.

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